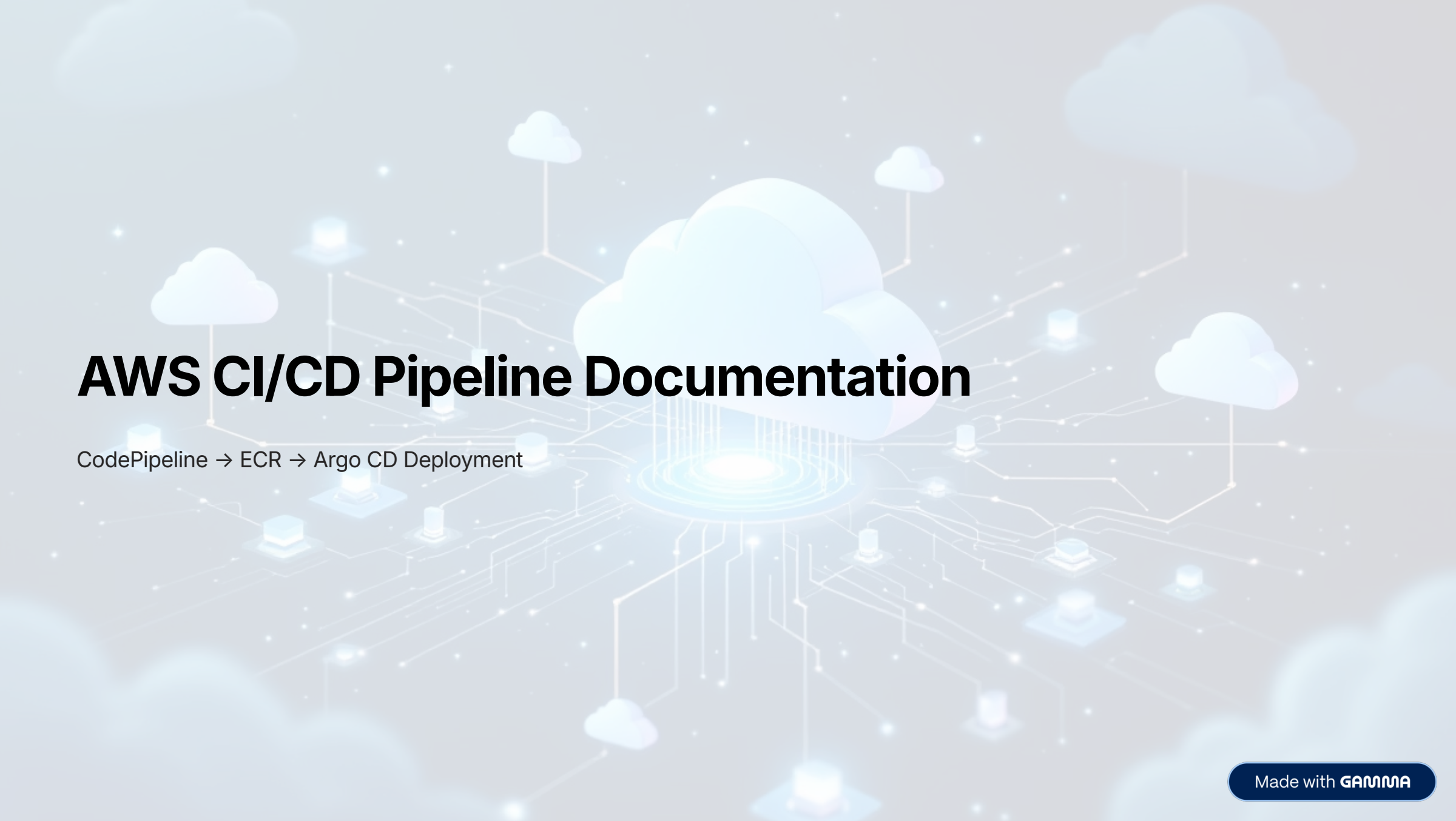


AWS CI/CD Pipeline Documentation



CodePipeline → ECR → Argo CD Deployment

Overview

This document explains the complete CI/CD pipeline for automated Kubernetes deployments. The workflow: Developer pushes code to GitHub → CodePipeline detects changes using GitHub triggers → Source stage pulls source code → Build stage (CodeBuild) builds Docker image and pushes to ECR → Argo CD automatically detects the new image version in ECR → Argo CD deploys the updated application to Kubernetes. This creates a fully automated, GitOps-style deployment process.

1. Pipeline Properties

Pipeline Type

V2

Execution Mode

SUPERSEDED (ensures only the latest pipeline run continues)

2. GitHub Trigger

Trigger Configuration:

- Trigger on: Push events
- Branch: shellkode-main
- Behavior: When code is pushed to the shellkode-main branch, the pipeline automatically starts

This ensures every commit triggers a fresh build and deployment cycle.

3. Source Stage

The Source stage is the first step in the pipeline:



This ensures the pipeline always builds from the most recent code commit.

4. Build Stage (CodeBuild → ECR)

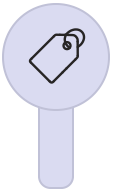
The Most Critical Stage



CodeBuild reads the Dockerfile from source code



Builds a Docker image for the application



Tags the image with the latest commit SHA or "latest"



Authenticates to Amazon ECR

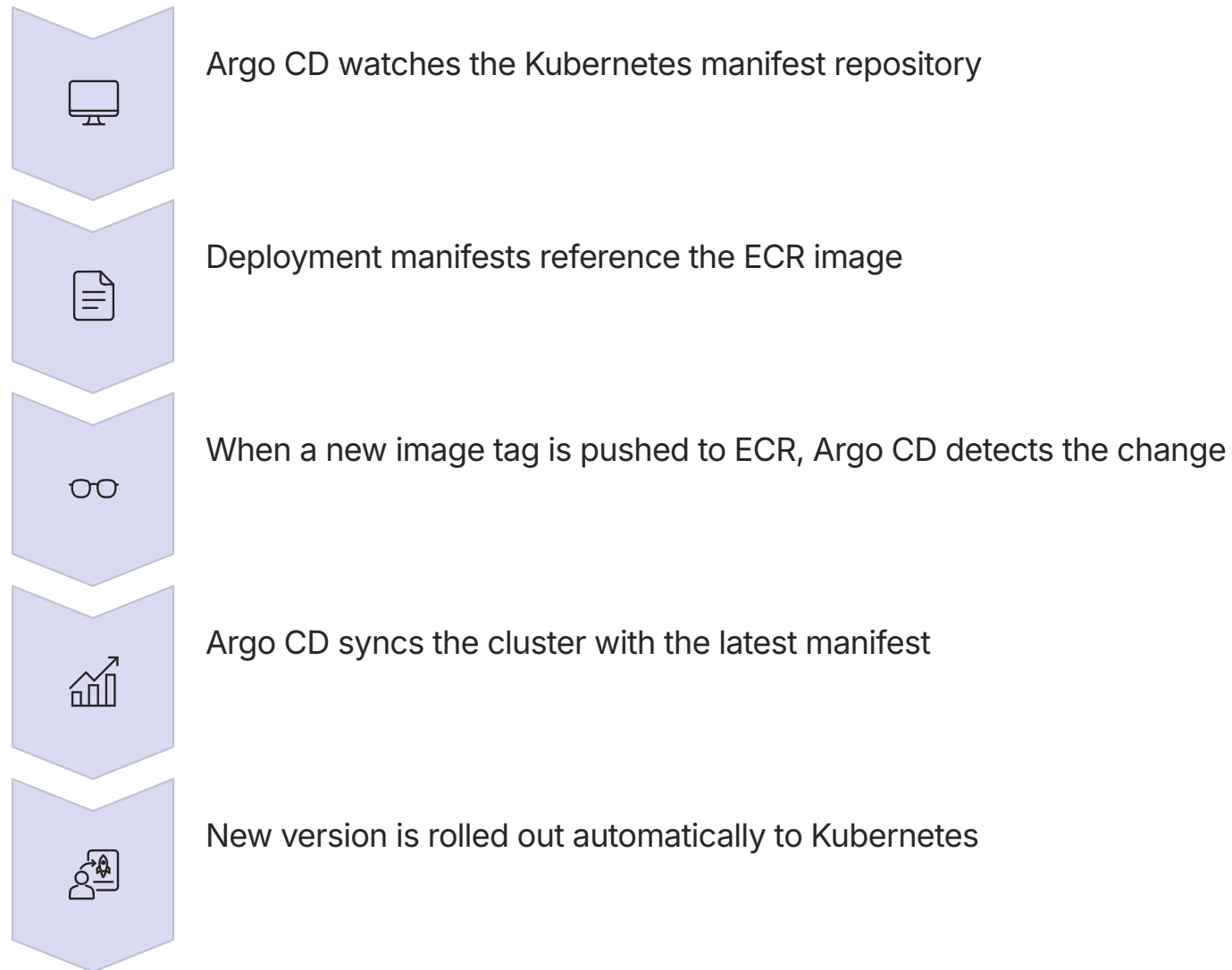


Pushes the Docker image to the ECR repository

This stage is fully automated and requires no manual intervention. The Docker image is now ready for deployment.

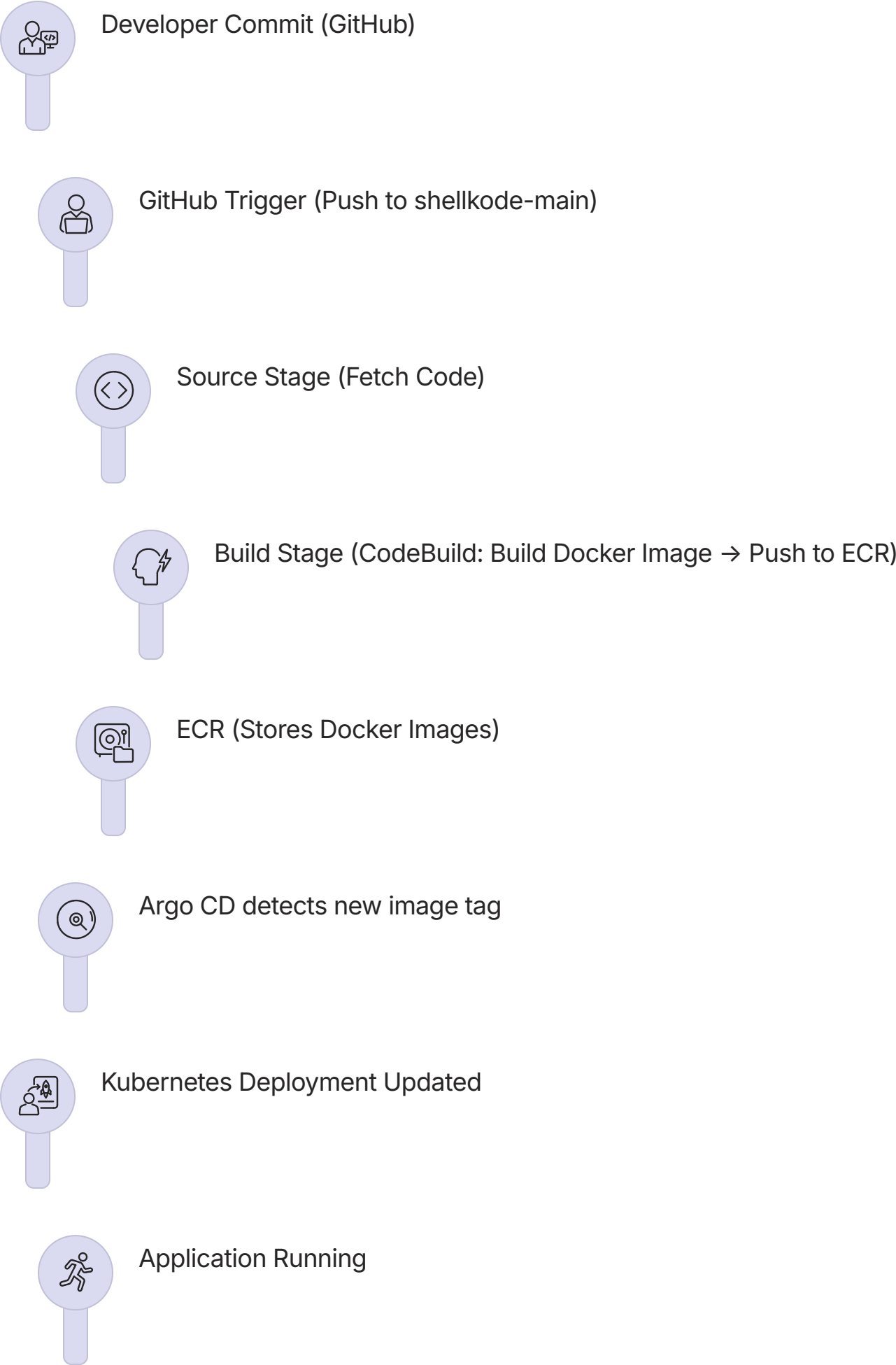
5. Deployment with Argo CD

After the Docker image is pushed to ECR, Argo CD takes over for continuous delivery:



This creates a GitOps-style deployment where the desired state is always defined in Git.

6. Full Architecture Diagram



7. Summary



Fully Automated CI/CD: Every commit triggers a complete build and deployment cycle



Docker-Based Build Process: Consistent, reproducible builds using CodeBuild



Secure Image Storage: Docker images stored securely in Amazon ECR



GitOps-Style Deployment: Argo CD ensures desired state is always defined in Git



Minimal Manual Intervention: Pipeline runs end-to-end without manual steps



Designed for Kubernetes: Scalable, cloud-native deployment architecture



Production-Ready: Reliable pipeline foundation for enterprise workloads